**CYBERSRS**

# Firewall Management System Requirements

Software Requirements Specification

DRAFT — human review required

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AI-generated advisory document. Review by a qualified professional is required.

Automated checks do not prove completeness or semantic correctness. SDLC suitability scores are rule-based points, not probabilities.

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# 1. Introduction

### Original project description

Design a centralized firewall management system for a university that manages firewall policies across academic, administrative, research, hostel, and guest networks. The system should allow authorized network administrators to create, review, update, and revoke firewall rules, enforce access restrictions between network segments, monitor blocked and allowed traffic, and generate alerts for suspicious or unauthorized communication attempts. The platform should maintain audit logs of firewall policy changes and administrative actions, provide a dashboard for reviewing network security events, and support multiple campus buildings. The solution should be secure, scalable, reliable, and manageable.

Summary: This specification records the project purpose, stakeholder constraints and unresolved decisions stated in the source evidence below.

### Items

### Stakeholder source statement E-001

Detail: Design a centralized firewall management system for a university that manages firewall policies across academic, administrative, research, hostel, and guest networks.

Status: confirmed

Evidence ids: E-001

### Stakeholder source statement E-002

Detail: The system should allow authorized network administrators to create, review, update, and revoke firewall rules, enforce access restrictions between network segments, monitor blocked and allowed traffic, and generate alerts for suspicious or unauthorized communication attempts.

Status: confirmed

Evidence ids: E-002

### Stakeholder source statement E-003

Detail: The platform should maintain audit logs of firewall policy changes and administrative actions, provide a dashboard for reviewing network security events, and support multiple campus buildings.

Status: confirmed

Evidence ids: E-003

### Stakeholder source statement E-004

Detail: The solution should be secure, scalable, reliable, and manageable.

Status: confirmed

Evidence ids: E-004

# 2. Functional requirements

Applicability reason: Independent source obligations; unspecified choices remain TBD.

### Requirements

### FR-001 - manage firewall policies across academic, administrative, research, hostel, and guest networks in a centralized system

Statement: The system shall manage firewall policies across academic, administrative, research, hostel, and guest networks in a centralized system.

Rationale: Source obligation E-001.

Priority: must

Dimensions: business; security

Basis: explicit

Evidence ids: E-001

### Acceptance criteria

Given: system manages firewall policies across academic, administrative, research, hostel, and guest networks in a centralized system

When: system initiates policy management for any network segment

Then: the requested firewall policy is applied and the resulting state is reflected in the network configuration

Method: test

Observable evidence: External test capture of the resulting response

### FR-002 - allow authorized network administrators to create firewall rules

Statement: The system shall allow authorized network administrators to create firewall rules.

Rationale: Source obligation E-002_O01.

Priority: should

Dimensions: business; security

Basis: explicit

Evidence ids: E-002

### Acceptance criteria

Given: authorized network administrators create firewall rules

When: the operation is initiated

Then: the system returns the newly created firewall rule in the active configuration state

Method: test

Observable evidence: External test capture of the resulting response

### FR-003 - allow authorized network administrators to review firewall rules

Statement: The system shall allow authorized network administrators to review firewall rules.

Rationale: Source obligation E-002_O02.

Priority: should

Dimensions: business; security

Basis: explicit

Evidence ids: E-002

#### Acceptance criteria

Given: authorized network administrators review firewall rules

When: the operation is initiated

Then: the system returns a complete list of current firewall rules with their status and scope

Method: inspection

Observable evidence: External test capture of the resulting state

### FR-004 - allow authorized network administrators to update firewall rules

Statement: The system shall allow authorized network administrators to update firewall rules.

Rationale: Source obligation E-002_O03.

Priority: should

Dimensions: business; security

Basis: explicit

Evidence ids: E-002

#### Acceptance criteria

Given: authorized network administrators update firewall rules

When: the operation is initiated

Then: the system applies the updated rule and reflects the change in the active configuration state

Method: analysis

Observable evidence: External test capture of the resulting state

### FR-005 - allow authorized network administrators to revoke firewall rules

Statement: The system shall allow authorized network administrators to revoke firewall rules.

Rationale: Source obligation E-002_O04.

Priority: should

Dimensions: business; security

Basis: explicit

Evidence ids: E-002

#### Acceptance criteria

Given: authorized network administrators revoke firewall rules

When: the operation is initiated

Then: the system removes the rule from active enforcement and updates the configuration state accordingly

Method: demonstration

Observable evidence: External test capture of the resulting state

### FR-006 - enforce access restrictions between network segments

Statement: The system shall enforce access restrictions between network segments.

Rationale: Source obligation E-002_O05.

Priority: should

Dimensions: business; security

Basis: explicit

Evidence ids: E-002

#### Acceptance criteria

Given: the system enforces access restrictions between network segments

When: a communication attempt is made across segments

Then: the system blocks or permits the traffic based on configured firewall rules

Method: test

Observable evidence: External test capture of the resulting response

### FR-007 - monitor blocked and allowed traffic

Statement: The system shall monitor blocked and allowed traffic.

Rationale: Source obligation E-002_O06.

Priority: should

Dimensions: business

Basis: explicit

Evidence ids: E-002

#### Acceptance criteria

Given: the system monitors blocked and allowed traffic

When: traffic flows occur across network segments

Then: The system monitors blocked and allowed traffic, and external test capture shows the resulting state of traffic flow across network segments.

Method: analysis

Observable evidence: External test capture of the resulting state

### FR-008 - generate alerts for suspicious or unauthorized communication attempts

Statement: The system shall generate alerts for suspicious or unauthorized communication attempts.

Rationale: Source obligation E-002_O07.

Priority: should

Dimensions: business; security

Basis: explicit

Evidence ids: E-002

### Acceptance criteria

Given: the system detects suspicious or unauthorized communication attempts

When: an anomalous traffic pattern is observed

Then: The system generates alerts for suspicious or unauthorized communication attempts when an anomalous traffic pattern is observed.

Method: demonstration

Observable evidence: External test capture of the resulting response

## FR-009 - maintain audit logs of firewall policy changes and administrative actions

Statement: The system shall maintain audit logs of firewall policy changes and administrative actions.

Rationale: Source obligation E-003_O01.

Priority: should

Dimensions: business; security; auditability; data

Basis: explicit

Evidence ids: E-003

### Acceptance criteria

Given: system maintains audit logs of firewall policy changes and administrative actions

When: the system records a change to firewall policy or an administrative action

Then: the audit log contains a timestamped entry with the action type, target resource, and actor identifier

Method: test

Observable evidence: External test capture of the resulting response

Unresolved decisions: Decision Q-3aae42e94793 is TBD.; Decision Q-64ef2e28335c is TBD.

## FR-010 - provide a dashboard for reviewing network security events

Statement: The system shall provide a dashboard for reviewing network security events.

Rationale: Source obligation E-003_O02.

Priority: should

Dimensions: business; data

Basis: explicit

Evidence ids: E-003

### Acceptance criteria

Given: system provides a dashboard for reviewing network security events

When: an administrator accesses the dashboard interface

Then: The administrator observes the dashboard interface displaying network security events, with output field set undecided.

Method: demonstration

Observable evidence: External test capture of the resulting state

### FR-011 - support multiple campus buildings

Statement: The system shall support multiple campus buildings.

Rationale: Source obligation E-003_O03.

Priority: should

Dimensions: business

Basis: explicit

Evidence ids: E-003

#### Acceptance criteria

Given: system supports multiple campus buildings

When: a user initiates a connection request from a building within the campus network

Then: The system successfully establishes a connection from a building within the campus network, with device approval criteria and implementation pending.

Method: test

Observable evidence: External test capture of the resulting state

## 3. Non-functional requirements

Applicability reason: Independent source obligations; unspecified choices remain TBD.

#### Requirements

### NFR-001 - be secure

Statement: The system shall be secure.

Rationale: Source obligation E-004_O01.

Priority: should

Dimensions: security

Basis: explicit

Evidence ids: E-004

#### Acceptance criteria

Given: A draft acceptance plan for the goal 'be secure'

When: An analyst reviews that acceptance plan

Then: The plan's measurable targets remain TBD or unconfirmed

Method: inspection

Observable evidence: External test capture of the inspection record

Unresolved decisions: Decision Q-7b2e5f89a0b1 is TBD.

### NFR-002 - be scalable

Statement: The system shall be scalable.

Rationale: Source obligation E-004_O02.

Priority: should

Dimensions: performance

Basis: explicit

Evidence ids: E-004

#### Acceptance criteria

Given: A draft acceptance plan for the goal 'be scalable'

When: An analyst reviews that acceptance plan

Then: The plan's measurable targets remain TBD or unconfirmed

Method: analysis

Observable evidence: External test capture of the resulting state

Unresolved decisions: Decision Q-543394346815 is TBD.

### NFR-003 - be reliable

Statement: The system shall be reliable.

Rationale: Source obligation E-004_O03.

Priority: should

Dimensions: availability

Basis: explicit

Evidence ids: E-004

#### Acceptance criteria

Given: A draft acceptance plan for the goal 'be reliable'

When: An analyst reviews that acceptance plan

Then: The plan's measurable targets remain TBD or unconfirmed

Method: inspection

Observable evidence: External test capture of the resulting response

Unresolved decisions: Decision Q-819bb986f419 is TBD.

### NFR-004 - be manageable

Statement: The system shall be manageable.

Rationale: Source obligation E-004_O04.

Priority: should

Dimensions: usability

Basis: explicit

Evidence ids: E-004

**Acceptance criteria**

Given: A draft acceptance plan for the goal 'be manageable'

When: An analyst reviews that acceptance plan

Then: The plan's measurable targets remain TBD or unconfirmed

Method: analysis

Observable evidence: External test capture of the inspection record

Unresolved decisions: Decision Q-5268b96e41b0 is TBD.

# 4. Security, privacy and compliance

Applicability reason: Existing canonical requirements, where applicable, are linked below; this section adds no separate obligation. Unresolved stakeholder decisions remain open.

Related requirement ids: FR-001; FR-002; FR-003; FR-004; FR-005; FR-006; FR-007; FR-008; FR-009; FR-010; FR-011; NFR-001

# 5. Data requirements

Applicability reason: Existing canonical requirements, where applicable, are linked below; this section adds no separate obligation. Unresolved stakeholder decisions remain open.

Related requirement ids: FR-009; FR-010; FR-011

# 6. Network and interfaces

Applicability reason: Existing canonical requirements, where applicable, are linked below; this section adds no separate obligation. Unresolved stakeholder decisions remain open.

Related requirement ids: FR-001; FR-002; FR-003; FR-004; FR-005; FR-006; FR-009; FR-010

# 7. Compliance and policy applicability matrix

# 8. Architecture

Summary: Proposed logical components and interactions mapped to canonical source obligations.

### Items

### Actor: authorized network administrators

Detail: Proposed logical endpoint for the source-stated actor: authorized network administrators.

Status: proposed

Evidence ids: E-002

Requirement ids: FR-002; FR-003; FR-004; FR-005

Component name: Actor: authorized network administrators

Responsibility: Represent the source-stated actor: authorized network administrators.

### Firewall Policy Management Engine

Detail: Proposed logical owner for FR-001, FR-006, FR-011.

Status: proposed

Evidence ids: E-001; E-002; E-003

Requirement ids: FR-001; FR-006; FR-011

Component name: Firewall Policy Management Engine

Responsibility: The system shall manage firewall policies across academic, administrative, research, hostel, and guest networks in a centralized system. The system shall enforce access restrictions between network segments. The system shall support multiple campus buildings.

### Rule Administration Interface

Detail: Proposed logical owner for FR-002, FR-003, FR-004, FR-005, NFR-002.

Status: proposed

Evidence ids: E-002; E-004

Requirement ids: FR-002; FR-003; FR-004; FR-005; NFR-002

Component name: Rule Administration Interface

Responsibility: The system shall allow authorized network administrators to create firewall rules. The system shall allow authorized network administrators to review firewall rules. The system shall allow authorized network administrators to update firewall rules. The system shall allow authorized network administrators to revoke firewall rules. The system shall be scalable.

### Traffic Monitoring and Alerting Module

Detail: Proposed logical owner for FR-007, FR-008, NFR-003, NFR-001.

Status: proposed

Evidence ids: E-002; E-004

Requirement ids: FR-007; FR-008; NFR-003; NFR-001

Component name: Traffic Monitoring and Alerting Module

Responsibility: The system shall monitor blocked and allowed traffic. The system shall generate alerts for suspicious or unauthorized communication attempts. The system shall be reliable. The system shall be secure.

### Audit Logging and Reporting Service

Detail: Proposed logical owner for FR-009, FR-010, NFR-004.

Status: proposed

Evidence ids: E-003; E-004

Requirement ids: FR-009; FR-010; NFR-004

Component name: Audit Logging and Reporting Service

Responsibility: The system shall maintain audit logs of firewall policy changes and administrative actions. The system shall provide a dashboard for reviewing network security events. The system shall be manageable.

## Data flows

### Proposed source-defined interaction

Detail: Proposed directed path for FR-002, FR-003, FR-004, FR-005.

Status: proposed

Evidence ids: E-002

Requirement ids: FR-002; FR-003; FR-004; FR-005

From component: Actor: authorized network administrators

To component: Rule Administration Interface

Data or action: The system shall allow authorized network administrators to create firewall rules. The system shall allow authorized network administrators to review firewall rules. The system shall allow authorized network administrators to update firewall rules. The system shall allow authorized network administrators to revoke firewall rules.

## Trust boundaries

### Proposed source-defined trust boundary

Detail: Proposed logical trust-domain crossing for FR-002, FR-003, FR-004, FR-005.

Status: proposed

Evidence ids: E-002

Requirement ids: FR-002; FR-003; FR-004; FR-005

Untrusted component: Actor: authorized network administrators

Trusted component: Rule Administration Interface

Enforcement: The system shall allow authorized network administrators to create firewall rules. The system shall allow authorized network administrators to review firewall rules. The system shall allow authorized network administrators to update firewall rules. The system shall allow authorized network administrators to revoke firewall rules.

# 9. Threats

### Assumptions

1. These are defensive planning hypotheses, not observed incidents or approved extra controls. Unrecognized threats and actual implementation effectiveness require human review.

# 10. Risks

# 11. User stories and use cases

### User stories and use cases

Id: UC-001

Actor: authorized network administrators

Goal: The system shall allow authorized network administrators to create firewall rules.

Story: As authorized network administrators, I need the following behavior: The system shall allow authorized network administrators to create firewall rules.

### Preconditions

1. authorized network administrators create firewall rules

### Steps

1. the operation is initiated

Success outcome: the system returns the newly created firewall rule in the active configuration state

Requirement ids: FR-002

Evidence ids: E-002

Id: UC-002

Actor: authorized network administrators

Goal: The system shall allow authorized network administrators to review firewall rules.

Story: As authorized network administrators, I need the following behavior: The system shall allow authorized network administrators to review firewall rules.

### Preconditions

1. authorized network administrators review firewall rules

### Steps

1. the operation is initiated

Success outcome: the system returns a complete list of current firewall rules with their status and scope

Requirement ids: FR-003

Evidence ids: E-002

Id: UC-003

Actor: authorized network administrators

Goal: The system shall allow authorized network administrators to update firewall rules.

Story: As authorized network administrators, I need the following behavior: The system shall allow authorized network administrators to update firewall rules.

### Preconditions

1. authorized network administrators update firewall rules

### Steps

1. the operation is initiated

Success outcome: the system applies the updated rule and reflects the change in the active configuration state

Requirement ids: FR-004

Evidence ids: E-002

Id: UC-004

Actor: authorized network administrators

Goal: The system shall allow authorized network administrators to revoke firewall rules.

Story: As authorized network administrators, I need the following behavior: The system shall allow authorized network administrators to revoke firewall rules.

### Preconditions

1. authorized network administrators revoke firewall rules

### Steps

1. the operation is initiated

Success outcome: the system removes the rule from active enforcement and updates the configuration state accordingly

Requirement ids: FR-005

Evidence ids: E-002

# 12. Verification and acceptance

### Tests

Id: TEST-001

Requirement ids: FR-001

### Procedure

1. Given: system manages firewall policies across academic, administrative, research, hostel, and guest networks in a centralized system
2. When: system initiates policy management for any network segment

Expected result: Expected: the requested firewall policy is applied and the resulting state is reflected in the network configuration

Evidence to record: Evidence: External test capture of the resulting response

Test type: positive

Id: TEST-002

Requirement ids: FR-002

### Procedure

1. Given: authorized network administrators create firewall rules
2. When: the operation is initiated

Expected result: Expected: the system returns the newly created firewall rule in the active configuration state

Evidence to record: Evidence: External test capture of the resulting response

Test type: positive

Id: TEST-003

Requirement ids: FR-003

### Procedure

1. Given: authorized network administrators review firewall rules
2. When: the operation is initiated

Expected result: Expected: the system returns a complete list of current firewall rules with their status and scope

Evidence to record: Evidence: External test capture of the resulting state

Test type: inspection

Id: TEST-004

Requirement ids: FR-004

### Procedure

1. Given: authorized network administrators update firewall rules
2. When: the operation is initiated

Expected result: Expected: the system applies the updated rule and reflects the change in the active configuration state

Evidence to record: Evidence: External test capture of the resulting state

Test type: analysis

Id: TEST-005

Requirement ids: FR-005

### Procedure

1. Given: authorized network administrators revoke firewall rules
2. When: the operation is initiated

Expected result: Expected: the system removes the rule from active enforcement and updates the configuration state accordingly

Evidence to record: Evidence: External test capture of the resulting state

Test type: positive

Id: TEST-006

Requirement ids: FR-006

### Procedure

1. Given: the system enforces access restrictions between network segments
2. When: a communication attempt is made across segments

Expected result: Expected: the system blocks or permits the traffic based on configured firewall rules

Evidence to record: Evidence: External test capture of the resulting response

Test type: positive

Id: TEST-007

Requirement ids: FR-007

### Procedure

1. Given: the system monitors blocked and allowed traffic
2. When: traffic flows occur across network segments

Expected result: Expected: The system monitors blocked and allowed traffic, and external test capture shows the resulting state of traffic flow across network segments.

Evidence to record: Evidence: External test capture of the resulting state

Test type: analysis

Id: TEST-008

Requirement ids: FR-008

### Procedure

1. Given: the system detects suspicious or unauthorized communication attempts
2. When: an anomalous traffic pattern is observed

Expected result: Expected: The system generates alerts for suspicious or unauthorized communication attempts when an anomalous traffic pattern is observed.

Evidence to record: Evidence: External test capture of the resulting response

Test type: positive

Id: TEST-009

Requirement ids: FR-009

### Procedure

1. Given: system maintains audit logs of firewall policy changes and administrative actions
2. When: the system records a change to firewall policy or an administrative action

Expected result: Expected: the audit log contains a timestamped entry with the action type, target resource, and actor identifier

Evidence to record: Evidence: External test capture of the resulting response

Test type: positive

Id: TEST-010

Requirement ids: FR-010

### Procedure

1. Given: system provides a dashboard for reviewing network security events
2. When: an administrator accesses the dashboard interface

Expected result: Expected: The administrator observes the dashboard interface displaying network security events, with output field set undecided.

Evidence to record: Evidence: External test capture of the resulting state

Test type: positive

Id: TEST-011

Requirement ids: FR-011

### Procedure

1. Given: system supports multiple campus buildings
2. When: a user initiates a connection request from a building within the campus network

Expected result: Expected: The system successfully establishes a connection from a building within the campus network, with device approval criteria and implementation pending.

Evidence to record: Evidence: External test capture of the resulting state

Test type: positive

Id: TEST-012

Requirement ids: NFR-001

### Procedure

1. Given: A draft acceptance plan for the goal 'be secure'
2. When: An analyst reviews that acceptance plan

Expected result: Expected: The plan's measurable targets remain TBD or unconfirmed

Evidence to record: Evidence: External test capture of the inspection record

Test type: inspection

Id: TEST-013

Requirement ids: NFR-002

### Procedure

1. Given: A draft acceptance plan for the goal 'be scalable'
2. When: An analyst reviews that acceptance plan

Expected result: Expected: The plan's measurable targets remain TBD or unconfirmed

Evidence to record: Evidence: External test capture of the resulting state

Test type: analysis

Id: TEST-014

Requirement ids: NFR-003

### Procedure

1. Given: A draft acceptance plan for the goal 'be reliable'

2. When: An analyst reviews that acceptance plan

Expected result: Expected: The plan's measurable targets remain TBD or unconfirmed

Evidence to record: Evidence: External test capture of the resulting response

Test type: inspection

Id: TEST-015

Requirement ids: NFR-004

**Procedure**

1. Given: A draft acceptance plan for the goal 'be manageable'
2. When: An analyst reviews that acceptance plan

Expected result: Expected: The plan's measurable targets remain TBD or unconfirmed

Evidence to record: Evidence: External test capture of the inspection record

Test type: analysis

Limitations: These verification plans have not been executed; their acceptance outcomes require testing.

# 13. Glossary

Summary: Source-present terminology; definitions introduce no additional product obligation.

**Items**

**Policy**

Detail: A rule or decision governing permitted actions; its project-specific meaning remains defined by the source requirements.

Status: proposed

Evidence ids: E-001; E-003

**Audit record**

Detail: A stored record of an event. Required event scope and retention are determined by the linked source, not this definition.

Status: proposed

Evidence ids: E-003

**Log**

Detail: A collection of event records. This term does not itself require additional logging fields, events or access permissions.

Status: proposed

Evidence ids: E-003

### Firewall

Detail: A control that permits or restricts network traffic according to rules; mentioning it does not authorize automatic changes.

Status: proposed

Evidence ids: E-001; E-002; E-003

# 14. SDLC recommendation and workflow

Selected model: Agile

Justification: The selection of Agile as the required model is based on a uniform score of 40 across all candidates, with no differentiating input evidence provided to favor one methodology over another. This lack of distinguishing factors—such as budget constraints, failure consequences, regulatory demands, or feedback cycles—means that no candidate model can be objectively ranked for suitability beyond equal point scores. The absence of evidence on critical tradeoffs (e.g., stability, schedule, uncertainty) renders all models equally viable in a vacuum. Given the requirement for centralized policy management, real-time rule operations, and continuous monitoring across diverse campus network segments, Agile supports iterative development, rapid feedback loops, and responsive adaptation—key to managing dynamic security needs. However, without explicit evidence on legacy systems, verification cadence, or regulatory compliance, this decision remains provisional. The existing requirements emphasize operational control, auditability, and real-time visibility, which align with Agile’s emphasis on incremental delivery and stakeholder collaboration. Nevertheless, the unknown factors such as schedule, uncertainty, and regulation prevent a definitive conclusion; thus, the work groups must be structured to accommodate iterative progress while ensuring no requirement duplication, per prior validation failure.

### Tradeoffs

1. Recurring delivery requires stakeholder review and backlog refinement; a cycle does not guarantee completeness or remove unresolved decisions.

### Phases

Name: Requirements and verification planning

Phase type: planning

### Activities

1. Review and baseline the source-defined work group: FR-001: The system shall manage firewall policies across academic, administrative, research, hostel, and guest networks in a centralized system.; FR-002: The system shall allow authorized network administrators to create firewall rules.; FR-003: The system shall allow authorized network administrators to review firewall rules.; FR-004: The system shall allow authorized network administrators to update firewall rules.; FR-005: The system shall allow authorized network administrators to revoke firewall rules.; FR-006: The system shall enforce access restrictions between network segments.; FR-007: The system shall monitor blocked and allowed traffic.; FR-008: The system shall generate alerts for suspicious or unauthorized communication attempts.; FR-009: The system shall maintain audit logs of firewall policy changes and administrative actions.; FR-010: The system shall provide a dashboard for reviewing network security events.; FR-011: The system shall support multiple campus buildings.
2. Review and baseline the source-defined work group: NFR-001: The system shall be secure.; NFR-002: The system shall be scalable.; NFR-003: The system shall be reliable.; NFR-004: The system shall be manageable.

3. Create the unit, integration, system and acceptance test plan before implementation; retain each authored acceptance criterion.

Roles: Proposed project owner; Proposed development lead; Proposed verification reviewer

### Deliverables

1. Versioned requirements baseline
2. Unit, integration, system and acceptance test plan
3. Unresolved-decision record

### Entry criteria

1. Authoritative inputs and canonical requirements are available for review.

### Exit criteria

1. A human reviewer has recorded the baseline decision and disposition of unresolved scope.

### Security checks

1. Review implementation and acceptance evidence against FR-001: The system shall manage firewall policies across academic, administrative, research, hostel, and guest networks in a centralized system.
2. Review implementation and acceptance evidence against FR-002: The system shall allow authorized network administrators to create firewall rules.
3. Review implementation and acceptance evidence against FR-003: The system shall allow authorized network administrators to review firewall rules.
4. Review implementation and acceptance evidence against FR-004: The system shall allow authorized network administrators to update firewall rules.
5. Review implementation and acceptance evidence against FR-005: The system shall allow authorized network administrators to revoke firewall rules.
6. Review implementation and acceptance evidence against FR-006: The system shall enforce access restrictions between network segments.
7. Review implementation and acceptance evidence against FR-008: The system shall generate alerts for suspicious or unauthorized communication attempts.
8. Review implementation and acceptance evidence against FR-009: The system shall maintain audit logs of firewall policy changes and administrative actions.
9. Review implementation and acceptance evidence against NFR-001: The system shall be secure.

Human approval: Pending human decision by Proposed project owner on this phase's deliverables and unresolved decisions; no approval is recorded by generation.

Requirement ids: FR-001; FR-002; FR-003; FR-004; FR-005; FR-006; FR-007; FR-008; FR-009; FR-010; FR-011; NFR-001; NFR-002; NFR-003; NFR-004

Name: Agile delivery cycle

Phase type: iteration

### Activities

1. Review and prioritize the backlog for the next iteration without changing source-defined scope.
2. Implement the source-defined work group: FR-001: The system shall manage firewall policies across academic, administrative, research, hostel, and guest networks in a centralized system.; FR-002: The system shall allow authorized network administrators to create firewall rules.; FR-003: The system shall allow authorized network administrators to review firewall rules.; FR-004: The

system shall allow authorized network administrators to update firewall rules.; FR-005: The systemshall allow authorized network administrators to revoke firewall rules.; FR-006: The system shallenforce access restrictions between network segments.; FR-007: The system shall monitor blockedand allowed traffic.; FR-008: The system shall generate alerts for suspicious or unauthorizedcommunication attempts.; FR-009: The system shall maintain audit logs of firewall policy changesand administrative actions.; FR-010: The system shall provide a dashboard for reviewing networksecurity events.; FR-011: The system shall support multiple campus buildings.

3. Implement the source-defined work group: NFR-001: The system shall be secure.; NFR-002: The system shall be scalable.; NFR-003: The system shall be reliable.; NFR-004: The system shall be manageable.
4. In an authorized isolated test environment, execute TEST-001: Given: system manages firewall policies across academic, administrative, research, hostel, and guest networks in a centralized system When: system initiates policy management for any network segment Expected: the requested firewall policy is applied and the resulting state is reflected in the network configuration
5. In an authorized isolated test environment, execute TEST-002: Given: authorized network administrators create firewall rules When: the operation is initiated Expected: the system returns the newly created firewall rule in the active configuration state
6. In an authorized isolated test environment, execute TEST-003: Given: authorized network administrators review firewall rules When: the operation is initiated Expected: the system returns a complete list of current firewall rules with their status and scope
7. In an authorized isolated test environment, execute TEST-004: Given: authorized network administrators update firewall rules When: the operation is initiated Expected: the system applies the updated rule and reflects the change in the active configuration state
8. In an authorized isolated test environment, execute TEST-005: Given: authorized network administrators revoke firewall rules When: the operation is initiated Expected: the system removes the rule from active enforcement and updates the configuration state accordingly
9. In an authorized isolated test environment, execute TEST-006: Given: the system enforces access restrictions between network segments When: a communication attempt is made across segments Expected: the system blocks or permits the traffic based on configured firewall rules
10. In an authorized isolated test environment, execute TEST-007: Given: the system monitors blocked and allowed traffic When: traffic flows occur across network segments Expected: The system monitors blocked and allowed traffic, and external test capture shows the resulting state of traffic flow across network segments.
11. In an authorized isolated test environment, execute TEST-008: Given: the system detects suspicious or unauthorized communication attempts When: an anomalous traffic pattern is observed Expected: The system generates alerts for suspicious or unauthorized communication attempts when an anomalous traffic pattern is observed.
12. In an authorized isolated test environment, execute TEST-009: Given: system maintains audit logs of firewall policy changes and administrative actions When: the system records a change to firewall policy or an administrative action Expected: the audit log contains a timestamped entry with the action type, target resource, and actor identifier
13. In an authorized isolated test environment, execute TEST-010: Given: system provides a dashboard for reviewing network security events When: an administrator accesses the dashboard interface Expected: The administrator observes the dashboard interface displaying network security events, with output field set undecided.
14. In an authorized isolated test environment, execute TEST-011: Given: system supports multiple campus buildings When: a user initiates a connection request from a building within the campus network Expected: The system successfully establishes a connection from a building within the campus network, with device approval criteria and implementation pending.

15. In an authorized isolated test environment, execute TEST-012: Given: A draft acceptance plan for the goal 'be secure' When: An analyst reviews that acceptance plan Expected: The plan's measurable targets remain TBD or unconfirmed
16. In an authorized isolated test environment, execute TEST-013: Given: A draft acceptance plan for the goal 'be scalable' When: An analyst reviews that acceptance plan Expected: The plan's measurable targets remain TBD or unconfirmed
17. In an authorized isolated test environment, execute TEST-014: Given: A draft acceptance plan for the goal 'be reliable' When: An analyst reviews that acceptance plan Expected: The plan's measurable targets remain TBD or unconfirmed
18. In an authorized isolated test environment, execute TEST-015: Given: A draft acceptance plan for the goal 'be manageable' When: An analyst reviews that acceptance plan Expected: The plan's measurable targets remain TBD or unconfirmed
19. Collect stakeholder feedback; agree its cadence with a human reviewer before scheduling cycles.
20. Return accepted stakeholder feedback and observed defects to the backlog and plan the next iteration.

Roles: Proposed project owner; Proposed development lead; Proposed verification reviewer

### Deliverables

1. Requirement-linked implementation increment
2. Recorded test evidence
3. Reviewed feedback and next-iteration backlog

### Entry criteria

1. Prior work, source scope and the reviewed test plan are available for iteration planning.

### Exit criteria

1. A human reviewer has assessed the increment, test evidence and feedback; unresolved changes return to the next iteration.

### Security checks

1. Review implementation and acceptance evidence against FR-001: The system shall manage firewall policies across academic, administrative, research, hostel, and guest networks in a centralized system.
2. Review implementation and acceptance evidence against FR-002: The system shall allow authorized network administrators to create firewall rules.
3. Review implementation and acceptance evidence against FR-003: The system shall allow authorized network administrators to review firewall rules.
4. Review implementation and acceptance evidence against FR-004: The system shall allow authorized network administrators to update firewall rules.
5. Review implementation and acceptance evidence against FR-005: The system shall allow authorized network administrators to revoke firewall rules.
6. Review implementation and acceptance evidence against FR-006: The system shall enforce access restrictions between network segments.
7. Review implementation and acceptance evidence against FR-008: The system shall generate alerts for suspicious or unauthorized communication attempts.
8. Review implementation and acceptance evidence against FR-009: The system shall maintain audit logs of firewall policy changes and administrative actions.

9. Review implementation and acceptance evidence against NFR-001: The system shall be secure.

Human approval: Pending human decision by Proposed project owner on this phase's deliverables and unresolved decisions; no approval is recorded by generation.

Requirement ids: FR-001; FR-002; FR-003; FR-004; FR-005; FR-006; FR-007; FR-008; FR-009; FR-010; FR-011; NFR-001; NFR-002; NFR-003; NFR-004

Name: Release and handoff

Phase type: release

### Activities

1. Review the exact requirements baseline, implementation links and verification evidence before a release decision.
2. Carry unresolved source decisions into handoff; generated text is not human approval.

Roles: Proposed project owner; Proposed development lead; Proposed verification reviewer

### Deliverables

1. Requirements and verification evidence package
2. Pending human release decision
3. Outstanding-decision record

### Entry criteria

1. Verification evidence and unresolved decisions are available for human review.

### Exit criteria

1. A human has recorded the release/handoff decision for the reviewed content.

### Security checks

1. Review implementation and acceptance evidence against FR-001: The system shall manage firewall policies across academic, administrative, research, hostel, and guest networks in a centralized system.
2. Review implementation and acceptance evidence against FR-002: The system shall allow authorized network administrators to create firewall rules.
3. Review implementation and acceptance evidence against FR-003: The system shall allow authorized network administrators to review firewall rules.
4. Review implementation and acceptance evidence against FR-004: The system shall allow authorized network administrators to update firewall rules.
5. Review implementation and acceptance evidence against FR-005: The system shall allow authorized network administrators to revoke firewall rules.
6. Review implementation and acceptance evidence against FR-006: The system shall enforce access restrictions between network segments.
7. Review implementation and acceptance evidence against FR-008: The system shall generate alerts for suspicious or unauthorized communication attempts.
8. Review implementation and acceptance evidence against FR-009: The system shall maintain audit logs of firewall policy changes and administrative actions.
9. Review implementation and acceptance evidence against NFR-001: The system shall be secure.

Human approval: Pending human decision by Proposed project owner on this phase's deliverables and unresolved decisions; no approval is recorded by generation.

Requirement ids: FR-001; FR-002; FR-003; FR-004; FR-005; FR-006; FR-007; FR-008; FR-009; FR-010; FR-011; NFR-001; NFR-002; NFR-003; NFR-004

Unknown factors: budget; delivery; failure_consequences; feedback; legacy; regulation; schedule; stability; uncertainty; verification

# SDLC rankings and decision evidence

Policy: sequential-sdlc-1.1

Generation profile: sequential-sdlc-generation-2.0

### Ranked candidates

Model: Agile

Score: 40

Reasons: No differentiating input evidence

Model: DevSecOps

Score: 40

Reasons: No differentiating input evidence

Model: Spiral

Score: 40

Reasons: No differentiating input evidence

Model: V-Model

Score: 40

Reasons: No differentiating input evidence

Model: Waterfall

Score: 40

Reasons: No differentiating input evidence

Model: Hybrid

Score: 40

Reasons: No differentiating input evidence

Model: Incremental

Score: 40

Reasons: No differentiating input evidence

# Evidence register

Id: E-001

Source: description

Quote: Design a centralized firewall management system for a university that manages firewall policies across academic, administrative, research, hostel, and guest networks.

Id: E-002

Source: description

Quote: The system should allow authorized network administrators to create, review, update, and revoke firewall rules, enforce access restrictions between network segments, monitor blocked and allowed traffic, and generate alerts for suspicious or unauthorized communication attempts.

Id: E-003

Source: description

Quote: The platform should maintain audit logs of firewall policy changes and administrative actions, provide a dashboard for reviewing network security events, and support multiple campus buildings.

Id: E-004

Source: description

Quote: The solution should be secure, scalable, reliable, and manageable.

# Source references

### kb-c1e0239af9dd46bd:1 - 03_OWASP_CheatSheets/107_OWASP_Zero_Trust_Architecture_Cheat_Sheet.md

Source document id: kb-c1e0239af9dd46bd

Chunk index: 1

Page or section: text chunk 2

Relevance score: 0.7085

Sha256: f5fe6d1e1cfefc05da1b1b9773acee6b7e9d156f4aa69ab0a14a7a1adfd2a8ef

### kb-c1e0239af9dd46bd:8 - 03_OWASP_CheatSheets/107_OWASP_Zero_Trust_Architecture_Cheat_Sheet.md

Source document id: kb-c1e0239af9dd46bd

Chunk index: 8

Page or section: text chunk 9

Relevance score: 0.7132

Sha256: f5fe6d1e1cfefc05da1b1b9773acee6b7e9d156f4aa69ab0a14a7a1adfd2a8ef

### kb-788cd8fba7ce0acb:3 - 03_OWASP_CheatSheets/083_OWASP_Network_Segmentation_Cheat_Sheet.md

Source document id: kb-788cd8fba7ce0acb

Chunk index: 3

Page or section: text chunk 4

Relevance score: 0.7096

Sha256: 2b91c4e103da8e53c98e3b6006b23c2bb02e8c55e5805a56f0068c7b31a477d8

### kb-c1e0239af9dd46bd:14 - 03_OWASP_CheatSheets/107_OWASP_Zero_Trust_Architecture_Cheat_Sheet.md

Source document id: kb-c1e0239af9dd46bd

Chunk index: 14

Page or section: text chunk 15

Relevance score: 0.7034

Sha256: f5fe6d1e1cfefc05da1b1b9773acee6b7e9d156f4aa69ab0a14a7a1adfd2a8ef

### kb-c1e0239af9dd46bd:9 - 03_OWASP_CheatSheets/107_OWASP_Zero_Trust_Architecture_Cheat_Sheet.md

Source document id: kb-c1e0239af9dd46bd

Chunk index: 9

Page or section: text chunk 10

Relevance score: 0.7019

Sha256: f5fe6d1e1cfefc05da1b1b9773acee6b7e9d156f4aa69ab0a14a7a1adfd2a8ef

### kb-788cd8fba7ce0acb:4 - 03_OWASP_CheatSheets/083_OWASP_Network_Segmentation_Cheat_Sheet.md

Source document id: kb-788cd8fba7ce0acb

Chunk index: 4

Page or section: text chunk 5

Relevance score: 0.7078

Sha256: 2b91c4e103da8e53c98e3b6006b23c2bb02e8c55e5805a56f0068c7b31a477d8

# Validation findings

Code: human_semantic_review_required

Severity: warning

Section: document

Message: Review completeness, actor permissions, contradictions and citation applicability; schema and link checks do not prove semantic correctness.

# Human review history

None recorded.

# AI drafting record

Requested model: cybersrs-qwen3-4b-ft:latest

Actual models used: cybersrs-qwen3-4b-ft:latest; qwen3:4b-instruct-2507-q4_K_M

### Section models

Introduction: cybersrs-qwen3-4b-ft:latest

Functional requirements: qwen3:4b-instruct-2507-q4_K_M

Non-functional requirements: qwen3:4b-instruct-2507-q4_K_M

Compliance and policy applicability matrix: cybersrs-qwen3-4b-ft:latest

Architecture: cybersrs-qwen3-4b-ft:latest

Threats: cybersrs-qwen3-4b-ft:latest

Risks: cybersrs-qwen3-4b-ft:latest

User stories and use cases: cybersrs-qwen3-4b-ft:latest

Verification and acceptance: cybersrs-qwen3-4b-ft:latest

Glossary: cybersrs-qwen3-4b-ft:latest

SDLC recommendation and workflow: qwen3:4b-instruct-2507-q4_K_M

Prompt version: sequential-1.32

# Requirements traceability matrix

| Requirement | Evidence / sources | Stories / tests / SDLC |
|---|---|---|
| FR-001 | E-001 | TEST-001, Requirements and verification planning, Agile delivery cycle, Release and handoff |
| FR-002 | E-002 | UC-001, TEST-002, Requirements and verification planning, Agile delivery cycle, Release and handoff |
| FR-003 | E-002 | UC-002, TEST-003, Requirements and verification planning, Agile delivery cycle, Release and handoff |
| FR-004 | E-002 | UC-003, TEST-004, Requirements and verification planning, Agile delivery cycle, Release and handoff |
| FR-005 | E-002 | UC-004, TEST-005, Requirements and verification planning, Agile delivery cycle, Release and handoff |
| FR-006 | E-002 | TEST-006, Requirements and verification planning, Agile delivery cycle, Release and handoff |
| FR-007 | E-002 | TEST-007, Requirements and verification planning, Agile delivery cycle, Release and handoff |
| FR-008 | E-002 | TEST-008, Requirements and verification planning, Agile delivery cycle, Release and handoff |
| FR-009 | E-003 | TEST-009, Requirements and verification planning, Agile delivery cycle, Release and handoff |
| FR-010 | E-003 | TEST-010, Requirements and verification planning, Agile delivery cycle, Release and handoff |
| FR-011 | E-003 | TEST-011, Requirements and verification planning, Agile delivery cycle, Release and handoff |
| NFR-001 | E-004 | TEST-012, Requirements and verification planning, Agile delivery cycle, Release and handoff |
| NFR-002 | E-004 | TEST-013, Requirements and verification planning, Agile delivery cycle, Release and handoff |
| NFR-003 | E-004 | TEST-014, Requirements and verification planning, Agile delivery cycle, Release and handoff |

| Requirement | Evidence / sources | Stories / tests / SDLC |
|---|---|---|
| NFR-004 | E-004 | TEST-015, Requirements and verification planning, Agile delivery cycle, Release and handoff |

Model: cybersrs-qwen3-4b-ft:latest + qwen3:4b-instruct-2507-q4_K_M; prompt: sequential-1.32